

Glycine Cleavage System (Glycine Decarboxylase Complex) in *Pseudomonas putida* KT2440 — Module/Pathway/Taxon Review

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00260 — Glycine, serine and threonine metabolism (GCS genes primarily bucketed under ppu00785) **Review focus:** Glycine cleavage system (GCS); generic module framed around human nonketotic hyperglycinemia (NKH)

1. Executive Summary

The glycine cleavage system (GCS, also called the glycine decarboxylase complex, GDC) is **fully satisfiable in the genome of *P. putida* KT2440**. All four catalytic/structural components — the P-protein (glycine decarboxylase), the H-protein (lipoyl-bearing carrier), the T-protein (aminomethyltransferase), and the L-protein (dihydrolipoyl dehydrogenase, shared E3) — are encoded. Notably, the P/H/T core is **duplicated** across two paralogous *gcv* operons: operon 1 at PP_0986–PP_0989 (*gcvT*-I, *gcvP*1, *gcvH*1, interleaved with a serine dehydratase) and operon 2 at PP_5192–PP_5194 (*gcvP*2, *gcvH*2, *gcvT*). The L-protein is not dedicated to the GCS; instead it is supplied *in trans* by one of three annotated dihydrolipoyl dehydrogenases (EC 1.8.1.4) that moonlight across the 2-oxoacid dehydrogenase complexes and the GCS. This architecture closely mirrors the *gcs1/gcs2* arrangement experimentally characterized in *Pseudomonas aeruginosa* PAO1.

The evidence base is predominantly **homology-based** (all *gcv* genes are UniProt PE3 "inferred from homology"; no strain-specific experimental characterization exists for KT2440 GCS proteins). The strongest transferable experimental support comes from the closely related *P. aeruginosa* PAO1, where the GCS is regulated by GcsR (PA2449), a TyrR-like enhancer-binding protein, and from *P. putida* itself for the shared E3/dihydrolipoamide dehydrogenase. We

identified a positional candidate regulator in KT2440 — a TyrR-like HTH protein PP_0997 adjacent to *gcv* operon 1 — and confirmed that the H-protein lipoylation machinery (lipA, lipB) is encoded.

For curation, the practical conclusions are: (i) mark the GCS module **covered** in KT2440; (ii) resolve L-protein **paralog ambiguity** (lpdG PP_4187 is the best-supported general E3, with lpd PP_5366 and lpdV PP_4404 as alternatives); (iii) keep the **choline/betaine → dimethylglycine → sarcosine → glycine** catabolic cluster (PP_0310–PP_0326) and serine hydroxymethyltransferase (glyA1/glyA2) **separate** as upstream glycine feeders, not GCS components; and (iv) flag the generic module as **needs_revision** because its "nonketotic hyperglycinemia / glycine encephalopathy" disease framing is a human inborn-error concept with no counterpart in a soil bacterium.

2. Target-Organism Pathway Definition

Included process. The GCS in *P. putida* KT2440 is the multienzyme system that catabolizes glycine by oxidative decarboxylation, coupling glycine breakdown to one-carbon (folate) metabolism. The net reaction is:



The reaction proceeds through a lipoyl-swinging-arm relay across four components:

1. **P-protein (GcvP; EC 1.4.4.2):** a PLP-dependent enzyme that decarboxylates glycine and transfers the residual aminomethyl group to the lipoyl arm of the H-protein.
2. **H-protein (GcvH):** a small, non-catalytic lipoylated carrier that shuttles the aminomethyl/methylene intermediate between components.
3. **T-protein (GcvT; EC 2.1.2.10):** releases ammonia from the aminomethyl-dihydrolipoyl-H-protein and transfers the one-carbon unit to tetrahydrofolate, forming 5,10-methylene-THF.
4. **L-protein (Lpd/DLD; EC 1.8.1.4):** reoxidizes the dihydrolipoyl H-protein, reducing NAD⁺ to NADH and regenerating the oxidized lipoyl arm. This flavoenzyme is shared with the pyruvate, 2-oxoglutarate, and branched-chain 2-oxoacid dehydrogenase complexes.

In a bacterium such as KT2440, the GCS is cytoplasmic (there are no mitochondria). Functionally, the system provides (a) a route for glycine catabolism / use of glycine as a carbon and nitrogen source, and (b) a major source of C1 units for folate-dependent biosynthesis. In

Pseudomonas, the duplicated operon (the regulated *gcs2*-type copy) is coupled with serine hydroxymethyltransferase and serine dehydratase to convert glycine → serine → pyruvate, feeding central metabolism.

Neighboring pathways to keep separate. - KEGG **ppu00260 (Glycine, serine and threonine metabolism)** is a broad *overview bucket*, not the GCS module. It contains threonine biosynthesis (*thrB*, *thrC*, *hom*), serine biosynthesis (*serA*, *serB*, *serC*), threonine/serine dehydratases, and many enzymes unrelated to glycine cleavage. The GCS proper is a distinct sub-module (KEGG maps the *gcv* genes to *ppu00785*). - **Choline/glycine-betaine catabolism (PP_0310–PP_0326)**: betaine → dimethylglycine → sarcosine → glycine. This *produces* glycine feeding into the GCS; it is upstream and lineage-specific, not part of the GCS. - **Serine hydroxymethyltransferase (glyA1 PP_0322, glyA2 PP_0671; EC 2.1.2.1)**: interconverts serine and glycine + 5,10-methylene-THF; a partner of, but distinct from, the GCS. - **Lipoate biosynthesis/attachment (lipA, lipB)**: required to activate the H-protein, but an accessory pathway.

Alternate names / database definitions. GCS = glycine decarboxylase complex (GDC) = glycine synthase (the reaction is reversible). Components: P-protein = GLDC/GcvP; H-protein = GCSH/GcvH; T-protein = AMT/GcvT; L-protein = DLD/Lpd/E3. The generic "nonketotic hyperglycinemia" label is a **human clinical framing** and should not be used as the module name for a bacterial taxon.

3. Expected Step Model

Step	Component	Activity (EC)	Expected in KT2440?	Status
S1	P-protein (GcvP)	glycine dehydrogenase (decarboxylating) 1.4.4.2	Yes	Covered (duplicated)
S2	H-protein (GcvH)	lipoyl-bearing carrier / lipoic acid binding	Yes	Covered (duplicated)
S3	T-protein (GcvT)	aminomethyltransferase 2.1.2.10	Yes	Covered (duplicated)
S4	L-protein (Lpd/DLD)	dihydrolipoyl dehydrogenase 1.8.1.4	Yes (shared, <i>in trans</i>)	Covered, paralog-ambiguous
A1	Lipoylation (LipA/LipB)	lipoyl synthase 2.8.1.8 / octanoyltransferase 2.3.1.181	Yes	Covered (accessory)
R1	Transcriptional regulator (GcsR-type)	TyrR-like EBP	Likely (positional candidate PP_0997)	Candidate_uncertain

All four core steps are encoded; the module is genome-supportable. The only genuine ambiguity is *which* dihydrolipoyl dehydrogenase serves the GCS H-protein.

4. Candidate Genes and Evidence

4.1 GCS core — operon 1 (PP_0986–PP_0989)

Gene	Locus	UniProt	Role	Evidence
gcvT-I	PP_0986	Q88P67	T-protein, aminomethyltransferase (EC 2.1.2.10)	Homology (PE3)
(tdcG-II)	PP_0987	Q88P66	L-serine dehydratase (EC 4.3.1.17) — <i>interleaved, not GCS</i>	Homology
gcvP1	PP_0988	Q88P65	P-protein, glycine dehydrogenase (EC 1.4.4.2)	Homology (PE3)
gcvH1	PP_0989	Q88P64	H-protein, lipoyl carrier	Homology (PE3)

The gene order (T–serine dehydratase–P–H) and the interleaved serine dehydratase parallel the *P. aeruginosa* *gcs2* operon (gcvH2–gcvP2–glyA2–sdaA–gcvT2), which forms a glycine → pyruvate pathway ([PMID: 27303730](#)).

4.2 GCS core — operon 2 (PP_5192–PP_5194)

Gene	Locus	UniProt	Role	Evidence
gcvP2	PP_5192	Q88CI9	P-protein 2 (EC 1.4.4.2)	Homology (PE3)
gcvH2	PP_5193	Q88CI8	H-protein 2	Homology (PE3)
gcvT	PP_5194	Q88CI7	T-protein (EC 2.1.2.10)	Homology (PE3)

A complete second P/H/T set. The duplication is a genuine feature and closely matches the two-operon organization documented in *P. aeruginosa*.

4.3 L-protein candidates (shared E3, EC 1.8.1.4)

Gene	Locus	UniProt	Evidence tier	Note
lpdG	PP_4187	Q88FB1	PE1 (protein level)	Strongest experimentally supported general/ GCS E3
lpd	PP_5366	Q88C17	PE3 (homology)	General E3 candidate
lpdV	PP_4404	Q88EP9	PE3 (homology)	Branched-chain 2-oxoacid dehydrogenase E3

No *gcv*-linked dedicated L-protein exists; the E3 is recruited *in trans*. In *P. putida*, the dihydrolipoamide dehydrogenase (E3) is experimentally established to serve both the α -keto acid dehydrogenase complexes and the GCS H-protein ([PMID: 15826505](#)). **lpdG (PP_4187, PE1)** is the best-supported general E3 and the leading GCS L-protein candidate; **lpdV** is most likely dedicated to the branched-chain complex.

4.4 Accessory and regulatory genes

Gene	Locus	UniProt	Role	Evidence
lipA	PP_4800	Q88DM5	Lipoyl synthase (EC 2.8.1.8)	Homology
lipB	PP_4801	Q88DM4	Octanoyltransferase (EC 2.3.1.181)	Homology
(lipA-2)	PP_4797	—	Second predicted lipoyl synthase	Prediction
glyA1	PP_0322	Q88R12	SHMT 1 (EC 2.1.2.1) — <i>partner, feeder</i>	Homology
glyA2	PP_0671	Q88Q27	SHMT 2 (EC 2.1.2.1) — <i>partner, feeder</i>	Homology
PP_0997 (candidate GcsR)	PP_0997	Q88P56	TyrR-like HTH regulator, adjacent to operon 1	Positional inference

4.5 Upstream glycine-feeder cluster (PP_0310–PP_0326) — NOT GCS

This cluster encodes the choline/glycine-betaine catabolic route: - **dgcA/dgcB (PP_0310/PP_0311)**: dimethylglycine dehydrogenase subunits (EC 1.5.8.-) - **PP_0312/PP_0313**: electron transfer flavoprotein α/β - **gbcA/gbcB (PP_0315/PP_0316)**: glycine-betaine demethylase/

dioxygenase (EC 1.13.-.-) - **ItaE (PP_0321)**: L-threonine aldolase (EC 4.1.2.48) - **soxBDAG (PP_0323–PP_0326)**: heterotetrameric sarcosine oxidase ($\alpha/\beta/\gamma/\delta$) - **PP_3775**: additional sarcosine oxidase (EC 1.5.3.1)

These convert betaine → dimethylglycine → sarcosine → glycine, generating the substrate that feeds the GCS. They belong to the *P. aeruginosa* GbdR regulon ([PMID: 24097953](#)) and are **upstream feeders**, not GCS components.

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 L-protein paralog ambiguity (highest-priority ambiguity)

Three EC 1.8.1.4 genes are annotated and none is physically linked to a *gcv* operon. The GCS must borrow one of them. Curation should not assign a single locus with certainty. Recommended interim assignment: **lpdG (PP_4187)** as the general/GCS E3 (PE1 support), with *lpd* and *lpdV* noted as alternatives. This is a **candidate_uncertain** designation for the specific gene, though the *step* is **covered**.

5.2 Duplicated P/H/T — which operon is "the" GCS?

Both operons appear complete. In *P. aeruginosa* the two copies are differentially regulated (one glycine-inducible via GcsR). In KT2440 the functional division of labor is unverified. Both should be recorded as covering the core steps; a note that duplication may reflect distinct regulatory/physiological roles is warranted.

5.3 Broad EC/GO and over-propagation risks

- **ppu00260 bucket over-inclusion**: The 66-gene candidate list is dominated by threonine/serine metabolism, betaine catabolism, phospholipid synthesis (*pssA*, *pcs*, PP_4677), and glyoxylate/glycerate enzymes that are *not* GCS. Many carry broad or partial EC numbers (e.g., EC 1.13.-.-, EC 1.1.1.-, EC 2.6.1.-). These should not be mapped to the GCS module.
- **"Betaine aldehyde dehydrogenase" duplications** (*betA/betB*/PP_0708) and multiple serine dehydratases (*tdcG-I/II/III*) reflect the overview-map nature of ppu00260, not GCS membership.
- **Regulator naming gap**: No gene is annotated *gcsR/gcvA/gcvR/gcvB* by name; PP_0997 is a positional inference only.

5.4 Disease-framing mismatch (module_needs_revision)

The generic module is framed around **nonketotic hyperglycinemia / glycine encephalopathy** — a human autosomal-recessive inborn error (GLDC ~80%, AMT ~20%, GCSH rare; [PMID: 33791923](#)). This clinical concept has **no counterpart in *P. putida***. The bacterial module should be re-scoped as "glycine cleavage / glycine catabolism and one-carbon provision," dropping the disease framing.

6. Module and GO-Curation Recommendations

Module step / element	Recommended status	Rationale
P-protein (S1)	covered	gcvP1 + gcvP2 encoded
H-protein (S2)	covered	gcvH1 + gcvH2 encoded; lipoylation machinery present
T-protein (S3)	covered	gcvT-I + gcvT encoded
L-protein (S4)	covered (step); candidate_uncertain (specific gene)	Shared E3 in trans; lpdG best candidate
Lipoylation accessory	covered	lipA/lipB present
GcsR-type regulator	candidate_uncertain	PP_0997 positional candidate only
Choline/ betaine→glycine cluster	not part of module (separate feeder module)	Upstream glycine source
Disease framing	module_needs_revision	NKH is human-specific; not applicable to bacteria

Module boundary corrections. - Split the module cleanly from the broad KEGG ppu00260 overview map. The GCS module should contain only P/H/T/L + lipoylation accessory (+ regulator). - Create or reference a **separate upstream module** for choline/glycine-betaine/sarcosine catabolism (the GbdR-type regulon) as the glycine-feeder route. - SHMT (glyA1/glyA2) and serine dehydratase should be represented as **partner/adjacent** steps, not GCS core.

GO considerations. Core GO annotations apply well: glycine dehydrogenase (decarboxylating) activity (GO:0004375), aminomethyltransferase activity (GO:0004047), dihydrolipoyl dehydrogenase activity (GO:0004148), lipoic acid binding, and glycine catabolic process (GO:0006546) / glycine cleavage. No new GO term requests appear necessary. A curation note should record that the L-protein GO annotation is shared/moonlighting and should not imply a dedicated GCS-only gene.

7. Genes to Promote to Full fetch-gene Review

1. **lpdG (PP_4187, Q88FB1)** — PE1 evidence; resolve whether it is the physiological GCS E3. Highest priority.
 2. **PP_0997 (Q88P56)** — candidate GcsR/TyrR-like regulator adjacent to *gcv* operon 1; confirm whether it activates the GCS operon.
 3. **gcvP1/gcvH1/gcvT-I (PP_0986–0989)** and **gcvP2/gcvH2/gcvT (PP_5192–5194)** — confirm operon structure, promoter architecture, and any functional divergence between the two copies.
 4. **lpd (PP_5366) and lpdV (PP_4404)** — clarify complex-specific assignments to rule in/out as GCS E3.
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8. Key Findings (Detailed)

F001 — A complete, duplicated GCS

UniProt/KEGG verification confirms all four GCS components are present. Operon 1 (PP_0986–PP_0989) carries *gcvT-I* (T), *gcvP1* (P), and *gcvH1* (H), interleaved with a serine dehydratase (PP_0987). Operon 2 (PP_5192–PP_5194) carries *gcvP2* (P), *gcvH2* (H), and *gcvT* (T). The shared L-protein is supplied in trans (*lpd*, *lpdG*, *lpdV*). All *gcv* genes are UniProt PE3 (inferred from homology) — there is no strain-specific experimental characterization. This mirrors the *P. aeruginosa gcs1/gcs2* arrangement in which the *gcs2* operon (*gcvH2-gcvP2-glyA2-sdaA-gcvT2*) is regulated by GcsR and converts glycine to pyruvate. The paper reports: "GcsR binds to an 18-bp consensus sequence (TGTAACG-N4-CGTTCCG) upstream of the *gcs2* operon, consisting of the *gcvH2*, *gcvP2*, *glyA2*, *sdaA*, and *gcvT2* genes" ([PMID: 27303730](#)).

F002 — L-protein is a shared, moonlighting subunit with paralog ambiguity

Three dihydrolipoyl dehydrogenase (EC 1.8.1.4) genes are annotated: *lpd* (PP_5366, PE3), *lpdG* (PP_4187, PE1 "evidence at protein level"), and *lpdV* (PP_4404, PE3). No *gcv*-linked dedicated L-protein exists. In *P. putida*, the E3 is experimentally established to serve both the 2-oxoacid dehydrogenase complexes and the GCS H-protein: "Dihydrolipoamide dehydrogenase (E3) catalyzes the reoxidation of dihydrolipoyl moiety of the acyltransferase components of three alpha-keto acid dehydrogenase complexes and of the hydrogen-carrier protein of the glycine cleavage system" (PMID: 15826505). *lpdV* is the branched-chain 2-oxoacid dehydrogenase E3; *lpdG* (PE1) is the strongest experimentally supported general/GCS E3 candidate.

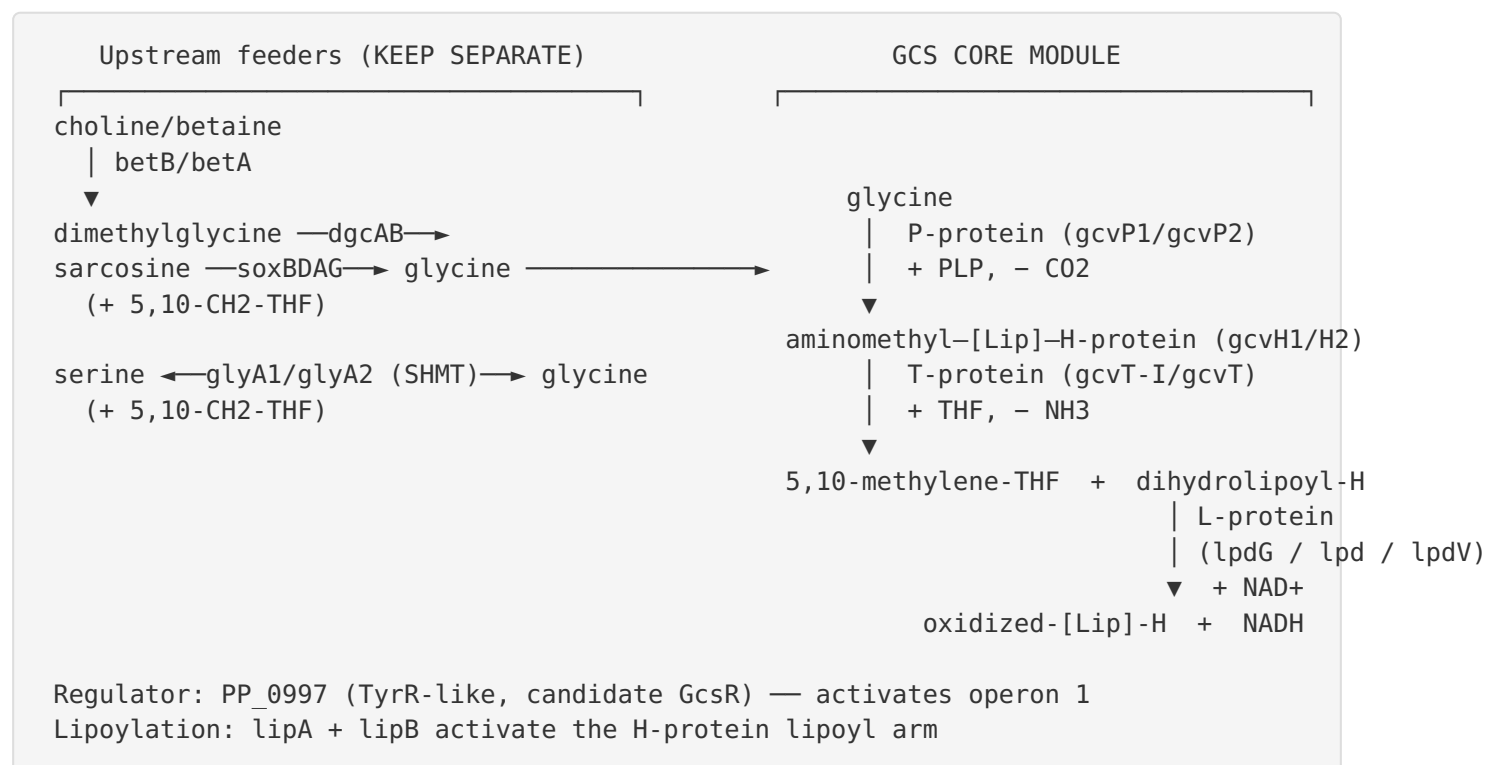
F003 — Betaine/sarcosine cluster is a lineage-specific glycine feeder; NKH framing not applicable

The PP_0310–PP_0326 cluster (*dgcAB*, ETF α/β , *gbcAB*, *ltaE*, *soxBDAG*) encodes the choline/glycine-betaine catabolic route (betaine $\rightarrow$ dimethylglycine $\rightarrow$ sarcosine $\rightarrow$ glycine) that generates glycine feeding into the GCS. In *P. aeruginosa* this is the GbdR regulon: "The GbdR regulon includes the genes encoding GB, dimethylglycine, sarcosine, glycine, and serine catabolic enzymes and the BetX and CbcXWV quaternary amine transport proteins" (PMID: 24097953). Heterotetrameric sarcosine oxidase ($\alpha\beta\gamma\delta$, matching *soxABGD*) "catalyzes the oxidation of sarcosine (N-methylglycine) to yield hydrogen peroxide, glycine and formaldehyde. In the presence of tetrahydrofolate, the oxidation of sarcosine is coupled to the formation of 5,10-methylenetetrahydrofolate" (PMID: 16820168). These are upstream feeder steps, not GCS components. The "nonketotic hyperglycinemia" framing is a human inborn error with no counterpart in *P. putida*.

F004 — Lipoylation machinery and a candidate regulator are present

The H-protein lipoylation machinery is encoded: *lipA* (PP_4800, lipoyl synthase, EC 2.8.1.8) and *lipB* (PP_4801, octanoyltransferase, EC 2.3.1.181), plus a second predicted lipoyl synthase (PP_4797). SHMT is duplicated (*glyA1*, *glyA2*), matching the two *gcv* operons. No gene is named *gcvA/gcvR/gcvB/gcsR*, but a TyrR-like HTH regulator PP_0997 lies immediately adjacent to operon 1, making it the strongest positional candidate for the GcsR-type activator: "GcsR belongs to a family of transcriptional regulators known as TyrR-like enhancer-binding proteins (EBPs) ... GcsR is the founding member of a new class of TyrR-like EBPs that function in the regulation of" glycine cleavage (PMID: 27303730).

9. Mechanistic Model / Interpretation



The KT2440 GCS is best understood as a **genomically redundant, regulator-controlled glycine catabolic hub**. Two complete P/H/T operons provide the enzymatic core; a single shared, moonlighting E3 closes the redox cycle; lipoylation enzymes activate the H-protein swinging arm; and a TyrR-like regulator (by analogy to *P. aeruginosa* GcsR) likely gates expression in response to glycine availability. The system is embedded in a larger glycine economy in which choline/betaine catabolism and SHMT supply glycine and one-carbon units. The human disease framing (NKH) is biologically irrelevant here and should be removed from the bacterial module.

10. Evidence Base

PMID	How it supports the review
27303730	Documents a duplicated <i>gcs2</i> operon (<i>gcvH2-gcvP2-glyA2-sdaA-gcvT2</i>) in <i>P. aeruginosa</i> and establishes GcsR as a TyrR-like enhancer-binding protein regulating the GCS — supports both the two-operon architecture and the PP_0997 regulator hypothesis in KT2440.
23457254	PA2449 (GcsR) is essential for glycine assimilation and controls a glycine cleavage system, SHMT, and serine dehydratase in <i>P. aeruginosa</i> — supports the coupled glycine→pyruvate module logic seen in KT2440 operon 1.
15826505	Establishes, using <i>P. putida</i> E3, that one shared dihydrolipoyl dehydrogenase serves the 2-oxoacid dehydrogenases and the GCS H-protein — direct support for the moonlighting L-protein.
24097953	Defines the GbdR regulon (dimethylglycine, sarcosine, glycine, serine catabolic enzymes) — supports treating PP_0310–PP_0326 as an upstream glycine feeder distinct from the GCS.
16820168	Defines the heterotetrameric sarcosine oxidase reaction producing glycine + 5,10-methylene-THF — supports the soxBDAG feeder assignment.
33791923	Describes human classic NKH genetics (GLDC/AMT) — confirms the disease concept is human-specific and does not transfer to bacteria.

Species-transfer assessment. Regulatory and paralog inferences derive from *P. aeruginosa* PAO1, a congener with strong but not identical physiology — transfer is **moderate-to-strong** for the two-operon architecture and GcsR concept, but the specific KT2440 regulator (PP_0997) and operon division of labor remain unverified. The E3 moonlighting evidence is from *P. putida* itself — transfer is **strong**. All KT2440 *gcv* gene calls are homology-based (PE3); no direct KT2440 experiments exist.

11. Limitations and Knowledge Gaps

1. **No direct KT2440 experiments.** Every *gcv* gene is UniProt PE3 (homology-inferred). Enzyme activities, operon boundaries, and regulation are predicted, not measured, in KT2440.
 2. **L-protein identity unresolved.** Which of *lpd*/*lpdG*/*lpdV* physiologically serves the GCS is not established for KT2440; only *lpdG* has protein-level evidence, and its complex specificity is not directly demonstrated for the GCS.
 3. **Regulator unconfirmed.** PP_0997 is a positional/homology candidate for a *GcsR*-type activator; no binding-site or expression data exist for KT2440.
 4. **Operon division of labor unknown.** The physiological roles/conditions distinguishing operon 1 vs. operon 2 are inferred from *P. aeruginosa*, not measured.
 5. **Module framing.** The commissioned module inherits a human disease scope that is inapplicable and could mis-guide bacterial curation if not revised.
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12. Proposed Follow-up Experiments / Actions (Curation)

1. **Promote *lpdG* (PP_4187) to full review** and cross-reference the 2-oxoacid dehydrogenase and GCS modules; annotate the L-protein step as "shared E3 (moonlighting)" rather than assigning a dedicated gene.
 2. **Promote PP_0997 to full review** as candidate *GcsR*; check for the *P. aeruginosa* *GcsR* 18-bp consensus (TGTAACG-N4-CGTTCCG) upstream of PP_0986.
 3. **Re-scope the module:** rename away from "nonketotic hyperglycinemia"; define scope as bacterial glycine cleavage / glycine catabolism + C1 provision. Mark `module_needs_revision`.
 4. **Split feeder module:** create/reference a separate choline-betaine-sarcosine → glycine catabolic module for PP_0310–PP_0326; keep it out of the GCS core.
 5. **Record duplication:** annotate both operons (PP_0986–0989 and PP_5192–5194) as covering P/H/T, with a note on potential regulatory divergence.
 6. **Experimental resolution (wet-lab, if pursued):** glycine-inducible RT-qPCR of both operons; Δ *lpdG*/ Δ *lpd*/ Δ *lpdV* complementation for GCS activity; Δ PP_0997 to test regulation; in vitro GCS reconstitution to confirm the physiological E3.
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13. Key References

- Sarwar Z. et al. *GcsR, a TyrR-Like Enhancer-Binding Protein, Regulates Expression of the Glycine Cleavage System in Pseudomonas aeruginosa PAO1.* [PMID: 27303730](#)
 - Lundgren B.R. et al. *Gene PA2449 is essential for glycine metabolism and pyocyanin biosynthesis in Pseudomonas aeruginosa PAO1.* [PMID: 23457254](#)
 - Asparagine-473 residue is important to the efficient function of human dihydrolipoamide dehydrogenase (P. putida E3 context). [PMID: 15826505](#)
 - Hampel K.J. et al. *Characterization of the GbdR regulon in Pseudomonas aeruginosa.* [PMID: 24097953](#)
 - *Heterotetrameric sarcosine oxidase: structure of a diflavin metalloenzyme at 1.85 Å resolution.* [PMID: 16820168](#)
 - *Genotypic and phenotypic features in Turkish patients with classic nonketotic hyperglycinemia.* [PMID: 33791923](#)
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Prepared as a species-aware module review to support manual satisfiability and gene-annotation curation. Direct KT2440 evidence is homology-based (UniProt PE3); mechanistic and regulatory inferences are transferred from P. aeruginosa (moderate-strong) and P. putida E3 studies (strong), with uncertainties flagged above.
